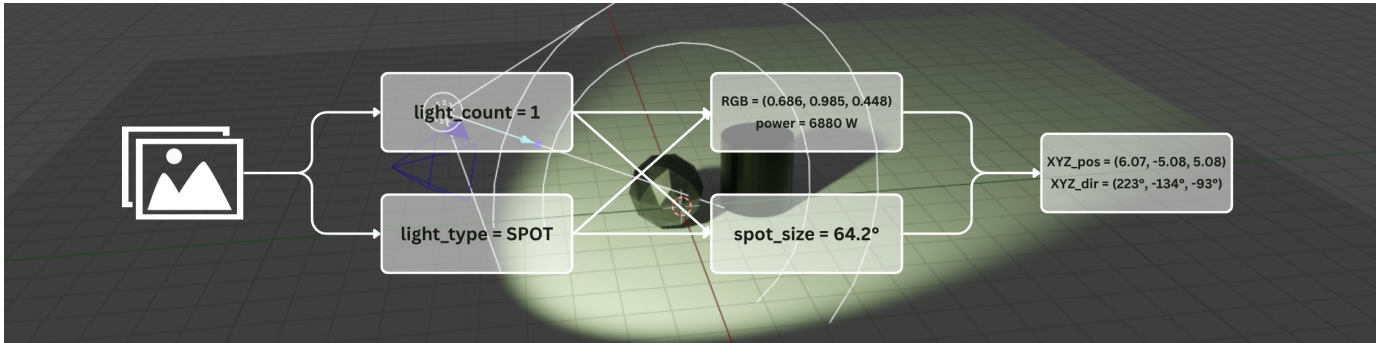


Inverse Rendering of Scene Lighting for Theatrical Lighting Reconstruction



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Abstract—This work investigates whether interpretable theatrical lighting parameters can be reconstructed from images using a synthetic inverse-rendering pipeline. The approach represents lighting as a small set of discrete sources with explicit attributes, including light type, count, position, angular direction, color, intensity, and spotlight cone angle. A modular prediction system is introduced that decomposes the problem into classification and regression sub-tasks, trained on a filtered dataset of 34,137 synthetic images rendered in Blender with ground-truth lighting labels. On a 500-image held-out test set, the system recovers basic lighting structure with almost perfect accuracy (100% and 99% accuracy on light type and light count, respectively) and geometric and photometric parameters with moderate error (e.g., 21.81° direction MAE and energy MAE of 2055.13). Results indicate that coarse lighting structure can be reliably recovered, while precise parameter estimation remains challenging due to ambiguity in shading and intensity cues. These findings demonstrate that modular inverse rendering is effective for structured lighting reconstruction in controlled settings, but highlight the need for improved supervision and real-world validation.

Index Terms—inverse rendering, machine learning, computer graphics, illumination modeling, lighting estimation, stage lighting

I. INTRODUCTION

This research project investigates whether interpretable theatrical lighting parameters can be reconstructed from images using a synthetic inverse-rendering pipeline that decomposes the problem into tractable sub-tasks such as light-type classification, light-count estimation, angular prediction, and color-power recovery.

Stage lighting design is extremely important for theatre and opera, not only because it provides illumination, but also because it functions as a primary storytelling tool [1]. Lighting shapes mood, guides the audience’s attention, and allows

designers to alter the perceived space of a performance in ways that influence how a scene is experienced. As Clifton Taylor, an internationally recognized theatrical scenic and lighting designer, argues, lighting can create “mood” and manipulate the audience’s sense of space and time [1]. As the technical complexity of contemporary theatre lighting continues to evolve, both its importance and its creative possibilities increase. Digital visualization tools and virtual models are increasingly being integrated into the creative and production workflows of large-scale theatre and opera, enabling teams to simulate performance spaces, scenic and lighting designs, and even aspects of production coordination and collaboration [2]. Because lighting and the ability to model that lighting plays such a significant artistic and operational role in modern theatre, the ability to reconstruct theatrical lighting is valuable not only for image-based analysis, but also for documentation, digital previsualization, production planning, and the preservation or remounting of performances.

However, reconstructing lighting from images is a fundamentally challenging problem. The formation of images depends on a combination of lighting, material properties, and geometry, making it difficult to determine a single correct solution. Multiple lighting configurations can produce similar results visually, especially when surfaces are primarily diffuse. Although highlights and shadows can provide useful cues, they are often incomplete or unclear in real images. Because of this, recovering accurate lighting usually requires making assumptions about the scene or learning patterns from data. Recent advances in machine learning and physically based rendering have made it possible to approach this problem using synthetic data. By generating rendered images with controlled variations in lighting, geometry, and materials, it becomes possible to train models with access to precise ground truth

lighting parameters from a renderer camera’s perspective. This synthetic inverse-rendering approach enables supervised learning of lighting estimation tasks while avoiding the ambiguity and noise present in real-world data. This is particularly useful for structured lighting setups, where configurations can be easily controlled and varied.

Instead of representing lighting as a continuous field or a latent variable, this work focuses on recovering lighting as a set of individual light sources. In theatrical lighting, scenes are typically built using a small number of spot or area lights, each with properties like position, direction, color, and intensity. Predicting lighting in this form makes it possible to directly reuse the results in real design and production workflows.

To do this, the problem is broken down into a series of smaller prediction tasks. These include identifying the type and number of lights, and then predicting properties for each light. The pipeline uses separate models for each task, with later predictions depending on earlier ones. This structure mirrors how lighting setups are actually built and makes the results easier to understand and use.

The contributions of this work include: 1.) a lighting reconstruction approach designed specifically for theatrical setups, 2.) a modular pipeline that breaks the problem into simpler, interpretable steps, and 3.) multiple synthetic datasets with controlled lighting variations and detailed ground truth for training and evaluation.

II. RELATED WORK

Prior work in inverse rendering has taken a variety of approaches to estimating lighting, differing mainly in how lighting is represented and what assumptions are made about the scene. Early analytical methods focused on understanding what information about lighting is actually recoverable from images. Ramamoorthi and Hanrahan show that diffuse reflectance limits the amount of lighting detail that can be captured, restricting recoverable illumination to low-frequency components [3]. This result helps explain why many methods struggle to recover detailed or directional lighting without additional cues.

To address this, image-based relighting methods incorporate stronger assumptions or additional signals. Yu et al. use shadow geometry and reference images to estimate light direction and intensity, enabling relighting without explicitly reconstructing the full structure of the scene [4]. These approaches demonstrate that useful lighting information can be extracted directly from images, but they often depend on controlled conditions or additional captured data.

More recent work uses neural networks to learn priors over lighting, geometry, and reflectance. Sengupta et al. predict these components from a single image, while NeRFactor models them jointly within a neural representation that supports relighting under novel conditions [5], [6]. Neural reflectance field methods extend this idea by directly modeling how appearance changes under different lighting, allowing high-quality relighting without explicitly recovering scene geometry [7]. While these approaches are effective for producing real-

istic images, they typically represent lighting in forms such as environment maps or implicit neural features, which are difficult to interpret or translate into discrete light sources.

Some work moves toward simpler and more interpretable lighting models. In augmented reality, Aittala uses a small number of directional lights and ambient terms to approximate real-world illumination for consistent rendering [8]. However, these simplified models are limited in their ability to represent more complex multi-light setups.

In parallel, synthetic datasets have become a common tool for training inverse rendering systems. By generating images with controlled variations in lighting and materials, these datasets provide exact ground truth for supervised learning [9]. This makes it possible to train models to predict lighting parameters directly, but most existing datasets and methods focus on general illumination rather than structured lighting configurations.

Across these approaches, a common trend is that lighting is either represented in a way that prioritizes visual realism, like environment maps and neural fields, or simplified for specific applications, such as directional light models. Neither approach directly matches how lighting is specified in domains such as theatre, where scenes are constructed using a small number of discrete lights with meaningful parameters.

This work builds on these ideas by focusing specifically on structured lighting reconstruction. Instead of predicting a global lighting representation, lighting is recovered as a set of individual light sources with interpretable properties. The problem is also decomposed into smaller steps, reflecting how lighting setups are designed in practice. This allows the predicted lighting to be directly used in downstream workflows, rather than only supporting relighting or rendering.

III. METHODOLOGY

This section describes the full inverse-lighting pipeline used in this project. The process starts with generating synthetic scenes in Blender with both randomized and systematically varied attributes, followed by metadata extraction and lighting label construction, and then training models for each task within a modular framework. The steps are presented in the order the data moves through the system so the pipeline is easier to follow and reproduce.

A. Synthetic Scene Generation

Synthetic data was used for training and testing primarily because it provides reliable ground truth lighting parameters, which are difficult to obtain from real images. Using Blender made it possible to directly access exact values for light position, direction, color, intensity, etc., at render time. Fig. 1 illustrates the controlled lighting setup used to generate ground truth parameters, including the spatial relationship between the camera, light source, and objects for data generation. Additionally, it allowed the use of a publicly available Blender-rendered lighting dataset [9] as a starting point, which provided a structured and accessible baseline for generating and organizing lighting variations.

From the render lighting dataset [9], scenes included a set of basic objects such as a cone, cube, cylinder, icosphere, sphere, and torus. For consistency and more obvious specular highlights, only the Plastic Glossy material was used from this dataset across experiments. The lighting setups considered included area, point, and spot lights, as well as tri-light configurations, with each lighting configuration rendered from 17 different camera viewpoints.

In addition, several custom datasets were created for this project using Blender sculpting and scripting to fill gaps in the data. The first consists of 2,313 rendered images of an icosphere with a Shiny Metal PBR material. Half of the images use a single spotlight sampled at 15-degree intervals around the object, while the other half use two spotlights with varied angles. This dataset is further divided into subsets for color, power, and spotlight size variations. The second dataset focuses on the color and power of a single spotlight and consists of 1,000 images of a Glossy sphere lit from a fixed position, with variations in the spotlight’s color and intensity. A final dataset introduces more geometric complexity than the single-object scenes used elsewhere. It contains renders of a Black Plastic cone and Silver Metallic cube placed together in the same scene under a single spotlight, allowing the system to capture cast shadows, inter-object occlusion, and multiple highlight patterns within a single image. This dataset includes 1,350 images with randomized variations in light angle, position, color, and intensity.

Together, there are 36,487 images available in the `/data` section of the codebase (see Fig. 2 for examples of the different datasets and lighting variations).

B. Lighting Parameterization

Instead of predicting a compressed or implicit representation of lighting, this system uses explicit, interpretable parameters. Each rendered image is paired with a structured set of labels that record lighting, camera, and scene information directly at render time. These values are used as the targets for all training and evaluation tasks.

For each image, the dataset includes full camera metadata, including position, forward/up/right basis vectors, and focal length. This makes it possible to express geometric quantities relative to the camera viewpoint. Each sample also records the

number of active lights and a fixed set of attributes for up to three lights.

Each light is described using a consistent set of parameters, including its type, energy, RGB color, world-space position, and both world-space and camera-space direction. Additional parameters are included when needed, such as spotlight cone angle and blend, or area light size and shape. To keep this representation consistent across scenes, the three brightest lights are recorded and sorted by energy before export so that each light occupies a stable slot (see Table I).

An important system decision was to use camera-space rep-

TABLE I
EXAMPLE ROW FROM THE TWO-OBJECT DATASET.

Parameter	Value
image_relpath	data/two_object/img_0123.png
num_active_lights	1
cam_pos_x, cam_pos_y, cam_pos_z	0.0, -3.0, 2.0
light0_type	SPOT
light0_energy	850.0
light0_color_r, g, b	1.0, 0.95, 0.8
light0_pos_x, y, z	1.2, -1.5, 2.5
light0_dir_cam_x, y, z	-0.42, 0.31, -0.85
light0_spot_cone_deg	35.0

resentations for geometric parameters. Although both world-space and camera-space values are available, early experiments showed that world-space predictions were sensitive to scene orientation and camera placement, which made reconstruction inconsistent. Projecting positions and directions into the camera coordinate frame reduces this issue and leads to more stable predictions.

The full label set is divided into groups that match the prediction pipeline. Geometric parameters (position and direction in camera space) are used for angular prediction, while photometric parameters (color and energy) are handled separately. Spotlight-specific parameters, such as cone angle and blend, are predicted by a dedicated model. For multi-light setups like tri-light configurations, the same structure is repeated across multiple ordered light slots.

Before training, direction vectors are normalized to unit length, and camera-space values are computed by projecting world-space quantities onto the camera basis. Energy is treated as a scalar regression target, and in some cases is log-transformed

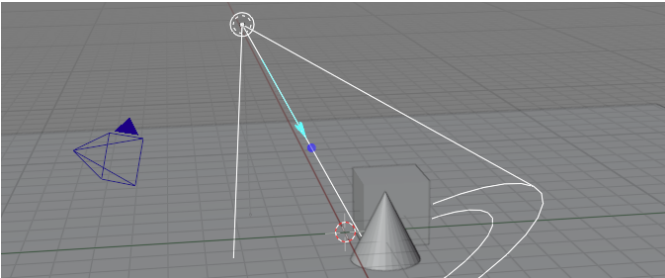


Fig. 1. Example Blender scene used for data generation. The setup shows the camera, spotlight, and object arrangement, along with the light direction and spot size.

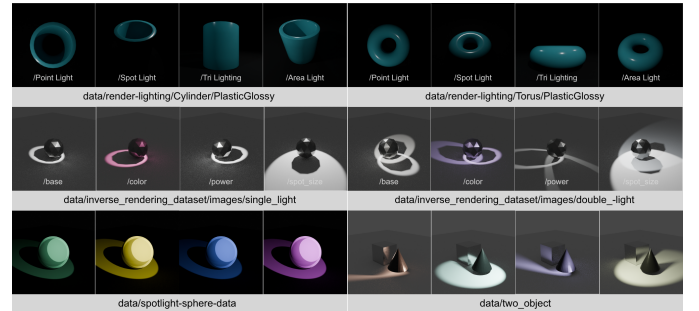


Fig. 2. Example images from the synthetic datasets, showing variation in object geometry, lighting configuration, and material properties.

before normalization to reduce instability from its large dynamic range.

Overall, this representation keeps the lighting parameters simple and interpretable while still capturing the information needed to reconstruct the scene in Blender.

C. Prediction Pipeline Models

The final prediction system uses a modular family of models rather than a single multitask network. This design was chosen due to instability observed in earlier attempts to jointly predict multiple lighting attributes. In practice, separating the tasks improves training stability and makes the system easier to evaluate and debug, since each model operates on a narrower and more consistent target distribution. It also allows failures to be isolated without retraining the entire pipeline.

At inference time, the system follows a staged process (Fig. 3). First, the image is analyzed to estimate scene structure, including the number of active lights and the lighting configuration type. These predictions are then used to route the input to specialized regression models. Finally, all outputs are merged into a single structured prediction record, producing a unified set of lighting estimates per image. This ordering mirrors the dataset organization and preserves interpretability throughout the pipeline.

The pipeline begins with two image-only classifiers operating on 224×224 RGB inputs. The `light_count_detector` is a shallow CNN with four convolutional blocks (32, 64, 128, 256 channels), followed by global average pooling and dense layers (128, 64), producing a softmax prediction over possible light counts. The `light_type_classifier` uses a similar backbone with a larger classification head (256, 128), and predicts one of five lighting classes: AREA LIGHT, DOUBLE_SPOT_SPOT, POINT LIGHT, SPOT LIGHT, or TRI LIGHTING. Together, these models provide a coarse structural description that defines downstream routing.

After scene structure is estimated, the pipeline switches to specialized regressors. Photometric properties are predicted using two image-only models operating on 128×128 RGB inputs. The `color_power_predictor` uses a three-layer CNN (32, 64, 128 channels, global average pooling, dense 128, dropout 0.3) with separate heads for RGB color and scalar energy. Energy is predicted in log space and converted back during inference. The `spot_size_predictor` follows a similar architecture and predicts the spotlight cone angle.

For geometric estimation, the `angular_predictor` handles single-spotlight scenes by predicting camera-space position and direction. It uses a hybrid architecture combining an image CNN (32, 64, 128, 256 channels with global average pooling and dense 128) and a small tabular branch (64 $\rightarrow$ 32). The output consists of six values corresponding to position and direction. The tabular branch is conditioned on compact metadata features, including camera information and previously predicted lighting attributes.

For multi-light configurations, the

`tri_angular_predictor` extends this approach to three lights, producing an 18-dimensional output vector containing position and direction for each light in camera space. It follows the same image-plus-tabular design, with conditioning inputs derived from the tri-light dataset.

Routing is performed as a final step based on predicted light count and type. The `angular_predictor`, `color_power_predictor`, and `spot_size_predictor` are used for single-spotlight cases, while the `tri_angular_predictor` is used for tri-light configurations. Outputs from all active models are combined into a single structured prediction per image.

This modular design improves stability and flexibility. Classification and regression tasks exhibit different optimization behavior, so separating them enables more effective tuning of preprocessing, normalization, and loss functions. It also allows individual modules to be retrained or replaced without affecting the rest of the system. From an analysis perspective, modularity simplifies failure diagnosis by isolating structural prediction errors from geometric or photometric errors. Most importantly, each module predicts explicit lighting parameters, preserving physical interpretability.

A final inference script integrates all components by loading trained models, applying routing logic, and assembling outputs into a unified record containing scene structure, geometry, and photometric estimates. This preserves flexibility during training while producing a single coherent prediction suitable for reconstruction.

D. Training & Preprocessing

All images were resized and normalized prior to training. Models operating on scene structure and geometry (`light_count_detector`, `light_type_classifier`, `angular_predictor`, and `tri_angular_predictor`) used 224×224 RGB inputs, while photometric models (`color_power_predictor` and `spot_size_predictor`) used 128×128 inputs. Pixel values were scaled to the $[0, 1]$ range.

For models that incorporate tabular inputs, categorical features were one-hot encoded and continuous values were standardized using training-set statistics. Direction vectors were normalized to unit length, and all geometric targets were expressed in camera space. Energy values were treated as scalar regression targets, and in photometric models were log-transformed prior to normalization to reduce instability from large dynamic range.

Each model was trained on a dataset tailored to its specific task. The `light_count_detector` was trained on a filtered dataset of 23,529 images, drawn primarily from the `render-lighting` dataset (21,216 samples) with additional coverage from the `inverse_rendering_dataset` (2,313 samples). This dataset was restricted to scenes with 1, 2, or 3 active lights, with a distribution of 17,385 single-light, 840 double-light, and

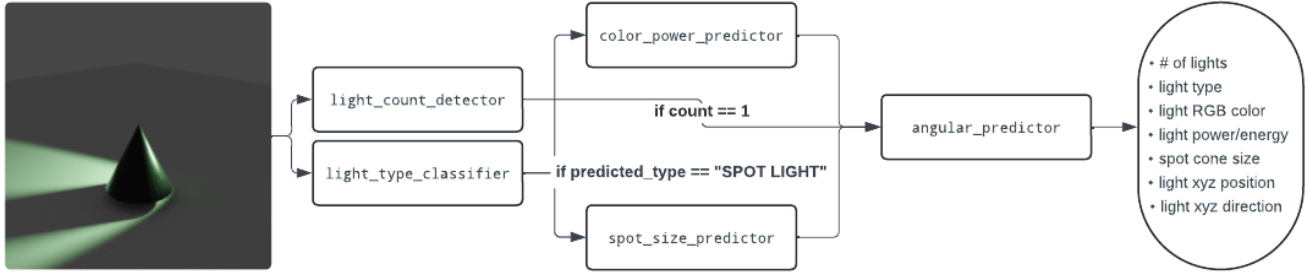


Fig. 3. Visualization of the model pipeline and data connections.

5,304 tri-light examples. The `light_type_classifier` was trained on a larger filtered dataset of 34,137 images. This dataset includes the same core sources but is not restricted to specific light counts, allowing the model to learn a broader range of lighting configurations beyond the constraints used for count prediction.

The `color_power_predictor` was trained on 2,350 images combining the `two_object` dataset (1,350 samples) and the controlled spotlight-sphere-data dataset (1,000 samples). These datasets were selected to isolate color and intensity variation while maintaining clear and consistent lighting effects. The `spot_size_predictor` was trained on a smaller dataset of 480 images from the `inverse_rendering_dataset`, focused specifically on variation in spotlight cone angle.

The `angular_predictor` was trained on 8,127 single-spotlight samples drawn from three sources: `render-lighting` (5,304), `inverse_rendering_dataset` (1,473), and `two_object` (1,350). The model uses a combined image and tabular input, where the tabular branch consists of 10 features including camera position, light metadata obtainable via other models, and a one-hot spotlight indicator.

The `tri_angular_predictor` was trained on a filtered subset of 1,326 images containing only tri-light configurations. These samples were restricted to consistent rendering conditions (PlasticGlossy material and a fixed rendering batch) to reduce variation unrelated to lighting geometry and improve stability when learning structured multi-light outputs.

All models were trained using standard supervised learning setups with train, validation, and test splits defined within each dataset. Care was taken to avoid data leakage, particularly for synthetic data where small changes in viewpoint or lighting can produce visually similar images.

In addition to quantitative evaluation, predictions were also validated qualitatively by reconstructing lighting setups in Blender using the predicted parameters. This provides a direct assessment of whether the models produce physically meaningful and visually consistent results, which is especially important since the primary goal is not exact numerical

accuracy, but general visual reconstruction.

IV. TESTING

A. Experimental Setup

Evaluation was performed using a fixed benchmark dataset of 500 held-out images (`test_data/test_metadata.csv`) generated in Blender. This dataset consists of a single Silver Metallic cone under a single spotlight, with controlled variation in lighting parameters including spotlight position, direction, cone angle (spot size), RGB color, and energy. Each image is paired with its corresponding ground-truth metadata, enabling direct comparison between predicted and true lighting parameters.

The test set was designed to isolate lighting effects while minimizing geometric complexity. By using a single object and material, variation in appearance is primarily driven by changes in lighting configuration rather than scene structure. This makes it possible to evaluate how well the models recover individual lighting attributes without confounding factors such as multi-object occlusion or complex geometry.

All evaluation was performed on model checkpoints trained on separate datasets, ensuring that the test set remained fully unseen during training. Predictions were generated using the full inference pipeline, including light-count and light-type estimation, followed by routing to the appropriate regression models. The outputs were then assembled into a prediction record row for each image.

Geometric predictions were evaluated in camera space to remain consistent with the training representation. No additional normalization or preprocessing was applied at test time beyond what was learned from the training data. This ensures that performance reflects the generalization ability of the trained models rather than adaptation to the test distribution.

B. Evaluation Metrics

Performance was evaluated using task-specific metrics aligned with the modular prediction pipeline. Because the benchmark set consists of 500 single-light SPOT scenes with controlled variation in lighting parameters, the evaluation emphasizes both geometric and photometric accuracy.

For geometric prediction, light direction was evaluated using angular error in degrees between predicted and ground-truth direction vectors. This is more interpretable than component-wise error for normalized direction outputs. Light position was evaluated in camera space using mean absolute error (MAE) per coordinate and aggregate Euclidean distance.

For photometric prediction, RGB color was evaluated using channel-wise MAE and overall RGB vector error. Light energy was evaluated with both absolute and relative error measures (MAE together with percentage-based metrics), since intensity spans a large dynamic range. Spotlight cone angle was evaluated using MAE and RMSE in degrees.

For completeness, the full pipeline evaluation also included classification metrics. Light type was evaluated with accuracy, and light count was evaluated with exact-match accuracy as well as error-based summaries such as MAE and RMSE, although the benchmark itself is dominated by single-light spotlight examples.

In addition to numerical metrics, qualitative evaluation was performed through manual visual comparison of predicted versus ground-truth outputs, including rendered comparison figures.

V. RESULTS

Table II presents the quantitative performance of the modular inverse-rendering pipeline on the 500-image test dataset. The system performs strongly on discrete scene interpretation tasks and moderately on continuous geometric and photometric regression. This pattern indicates that coarse lighting structure is easier to recover than precise physical parameters, particularly when those parameters depend on subtle shading cues, viewpoint, or dynamic range. Figure 4 provides a qual-

TABLE II
QUANTITATIVE RESULTS ON THE 500-IMAGE TEST DATASET.

Task	Metric	Value
<i>Classification</i>		
Light Type	Accuracy	100%
Light Count	Exact Accuracy	99.6%
Light Count	MAE	0.004
<i>Geometry</i>		
Direction	Angular MAE	21.81°
Direction	Angular RMSE	32.62°
Position	MAE (x, y, z)	1.20, 0.61, 0.76
Position	Mean L2 Error	1.72
<i>Photometric</i>		
Color (RGB)	MAE (R, G, B)	0.138, 0.134, 0.148
Color	Mean L2 Error	0.261
Energy	MAE	2055.13
Energy	R^2	0.225
<i>Spotlight Geometry</i>		
Spot Size	MAE	10.55°
Spot Size	RMSE	13.08°
<i>Overall</i>		
Pipeline Score	Aggregate Score	86.35
Coverage	Prediction Coverage	99.89%

itative comparison between ground-truth and reconstructed lighting. The model consistently recovers overall illumination

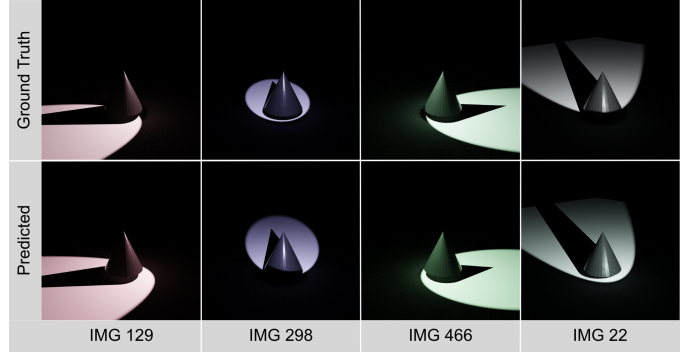


Fig. 4. Qualitative comparison between ground-truth and predicted lighting configurations.

structure, including color, shadow placement, and general light direction. However, visible differences in highlight sharpness, shadow boundaries, and intensity falloff reflect errors in geometric and photometric prediction. A representative failure case is visible in the fourth column of Figure 4, where the predicted RGB deviates noticeably from the ground truth. Although the overall lighting configuration remains plausible, the color shift alters the perceived material response and highlight tone, illustrating that the model captures global illumination structure but struggles to separate color and intensity under strong specular interactions.

A. Best-Performing Tasks

The strongest results were obtained on light-type and light-count prediction, achieving 100% and 99.6% accuracy respectively. These results indicate that the pipeline reliably recovers scene structure, enabling correct routing of inputs to downstream models.

Among regression tasks, spotlight cone angle and RGB color prediction were comparatively stable. Spot size achieved a mean absolute error of 10.55°, while RGB MAE values were 0.138, 0.134, and 0.148. These results suggest that mid-level photometric attributes can be recovered with reasonable accuracy, although exact reproduction remains imperfect.

As shown in Figure 5, spotlight cone prediction follows the overall trend of the ground truth but exhibits noticeable variance, particularly at larger angles. Task-specific evaluation showed that some regressors perform better in isolation than within the full pipeline. For example, the spot-size predictor achieved 6.08° MAE and 9.04° RMSE on its held-out test set, suggesting that some of the error increase in the full benchmark may be due to pipeline integration effects rather than the regressor alone.

B. Hardest Tasks

The most challenging tasks were light energy and angular prediction. Energy prediction produced an MAE of 2055.13 with low explanatory power ($R^2 = 0.225$), reflecting the difficulty of modeling high dynamic range intensity. As shown in Figure 6, predictions are compressed toward a narrower range and deviate from the ideal line.

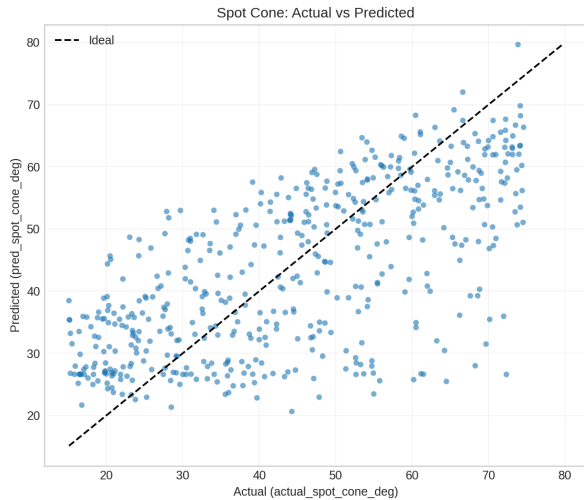


Fig. 5. Predicted vs. ground truth spotlight cone angles.

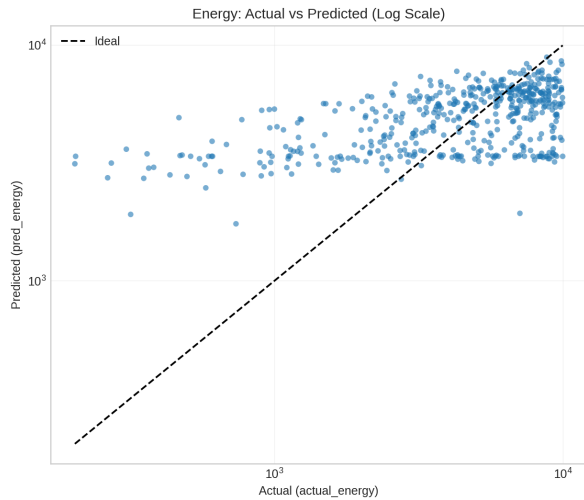


Fig. 6. Predicted vs. ground truth light energy (log scale).

Geometric prediction also remains difficult. Direction error reached 21.81° , indicating that the model often predicts plausible but not exact orientations. These errors are sufficient to produce visually reasonable reconstructions but can lead to noticeable discrepancies in lighting behavior.

VI. DISCUSSION

The results demonstrate that the modular inverse-rendering pipeline is effective at recovering coarse lighting structure, particularly for discrete tasks such as light type and count prediction. Near-perfect classification accuracy, including a 98% F1 accuracy in varied light type-only testing, indicates that the system reliably interprets high-level scene structure, which is essential for downstream parameter estimation. In contrast, continuous geometric and photometric predictions have higher error, reflecting the inherent ambiguity of inferring lighting parameters from image-based cues. Direction and po-

sition must be inferred from shading gradients, shadows, and specular highlights, which can vary significantly with small changes in viewpoint or material response. Energy prediction is particularly difficult due to its large dynamic range and the ambiguity between brightness, distance, color, and exposure. Qualitative results support this interpretation. In many cases, reconstructed lighting appears visually plausible even when numerical errors are present, suggesting that the model captures perceptually meaningful approximations. However, failure cases demonstrate that deviations in color, intensity, or direction can lead to noticeable differences in appearance, especially under high-contrast or focused lighting.

The modular design of the pipeline provides both strengths and limitations. By separating classification and regression tasks, the system avoids forcing a single model to learn incompatible objectives. However, this design introduces dependencies between stages: errors in light count or type prediction can propagate to downstream regressors and degrade overall performance. When upstream predictions are incorrect, downstream models may receive incompatible inputs, resulting in compounding errors in the final reconstruction. This cascading failure mode is an inherent limitation of staged pipelines and highlights the need for future work on uncertainty-aware routing or joint refinement strategies.

A central limitation of this work is the reliance on synthetic data. While synthetic rendering provides precise ground-truth labels and controlled variation, it does not fully capture the complexity of real-world theatrical environments. As a result, this system has not been validated on real stage imagery, and its performance in real productions remains unknown. Therefore, it's expected for there to be a gap between synthetic and real-world performance.

Overall, these results indicate that inverse rendering from synthetic data is effective for recovering interpretable lighting structure in controlled settings. However, the gap between structural understanding and precise parameter recovery remains significant, particularly for geometry and intensity.

VII. CONCLUSION

This work investigates the problem of reconstructing interpretable theatrical lighting parameters from images. Instead of representing lighting as a latent or global signal, the proposed approach models lighting as a small set of discrete sources with physically meaningful attributes.

A modular inverse-rendering pipeline was developed using synthetic datasets generated in Blender. The system decomposes the task into scene-structure prediction and specialized regression modules, producing a unified and interpretable lighting estimate for each image.

The results show that high-level lighting structure can be recovered reliably, while precise estimation of continuous parameters such as energy and exact geometry remains challenging. These findings demonstrate that modular inverse rendering is a viable approach for structured lighting reconstruction in controlled environments.

Future work will focus on improving precision and gener-

alization through more diverse datasets, stronger geometric supervision, and evaluation on real-world stage imagery. Extending the system to more complex multi-light scenes and reducing sensitivity to photometric ambiguity are key steps toward practical deployment.

ACKNOWLEDGMENT

A huge thank you to Dr. Joe Geigel for his guidance and assistance throughout this project via the XR Live Vertically Integrated Project program.

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APPENDIX A
CODE REPOSITORY

<https://github.com/kaclark219/inverse-rendering>

APPENDIX B
LOG OF WORK COMPLETED

TABLE III
WORK TIMELINE (HOURS + TASK SUMMARY)

Week	Daily Work
Jan 19–25	Mon: - Tue: 1h (Lighting kickoff meeting) Wed: - Thu: - Fri: 3h (Literature review) Sat: 3.5h (Literature review cont.) Sun: 0.25h (Literature summary)
Jan 26–Feb 1	Mon: 0.5h (Data workplan) Tue: - Wed: 3h (Blender data extraction) Thu: - Fri: - Sat: 3h (Data formatting export) Sun: 4h (Data preprocessing + baseline CNN)
Feb 2–8	Mon: 1.5h (Dataset fixes) Tue: 3h (Refactoring) Wed: - Thu: 2.25h (New training pipeline) Fri: - Sat: - Sun: 2.5h (Visual validation/testing)
Feb 9–15	Mon: - Tue: 3h (Model resets) Wed: - Thu: - Fri: 2.75h (Evaluation + visual simulation) Sat: - Sun: 3h (Color/energy dataset modeling)
Feb 16–22	Mon: 2h (Dataset creation) Tue: - Wed: 1h (Dataset creation cont.) Thu: - Fri: - Sat: 2h (Dataset creation cont.) Sun: 2h (Color/energy training)
Feb 23–Mar 1	Mon: 1.5h (Tri-light angular model) Tue: 1.5h (Tri-light angular model cont.) Wed: - Thu: - Fri: 4h (Light count/type prediction) Sat: - Sun: 2h (Full-system testing)
Mar 2–8	Mon: - Tue: 1.5h (Documentation formatting) Wed: - Thu: - Fri: - Sat: 4h (Additional data modeling + rendering) Sun: -
Mar 9–15 (Spring Break)	Mon: - Tue: 2h (Merging datasets) Wed: - Thu: - Fri: 3.5h (Light classification additions) Sat: - Sun: 2h (Fine-tuning)
Mar 16–22	Mon: 3.5h (Spot-size prediction/testing) Tue: 1h (Angular dependencies) Wed: 1.75h (Data flow fixes) Thu: - Fri: - Sat: - Sun: 4h (Large-scale testing)
Mar 23–29	Mon: - Tue: 2h (Introduction writing) Wed: 1h (Related works writing) Thu: 1.5h (Related works writing cont.) Fri: - Sat: 5.5h (More training data + training) Sun: -
Mar 30–Apr 5	Mon: (Training + evaluation fixes) Tue: 1h (Methods writing) Wed: 1h (Methods writing cont.) Thu: 1h (Methods writing cont.) Fri: 0.5h (Methods writing cont.) Sat: 1h (Methods writing cont.) Sun: -
Apr 6–12	Mon: 3.5h (Results writing + rendering comparisons) Tue: - Wed: 2h (Discussion writing) Thu: 1.5h (Paper visuals creation) Fri: - Sat: 1.5h (Conclusion writing) Sun: 0.75h (Abstract writing)
Apr 13–19	Mon: - Tue: 3h (Draft editing) Wed: 1.5h (Converting handwritten work notes to table) Thu: - Fri: - Sat: 1.5h (Header diagram creation) Sun: -